Empirical and Formal Verification of Boyle's Law with a Machine-Checked Least-Squares Analysis

Anonymous Authors[1]

Abstract

We verify Boyle's law—that the product of pressure and volume of a gas is constant under isothermal conditions—against a set of 12 laboratory measurements at $T = 293.15$ K. After removing a single transcription outlier, the product pV is 249.3 ± 2.6 kPa·L (relative spread 1.06%), and a least-squares fit of the inverse relation $p = C/V$ yields $C = 248.9$ kPa·L with $R^2 = 0.9993$. We complement the empirical analysis with a machine-checked formalization in the Lean 4 theorem prover: we prove the derivative of the least-squares objective, the normal equation for regression through the origin, the convexity of the objective, and the inverse-proportionality statement of Boyle's law. The results confirm the law within experimental uncertainty and demonstrate a reproducible experiment–verification–report pipeline.

1. Introduction

Boyle's law states that, at constant temperature, the pressure p and volume V of a fixed amount of gas are inversely proportional, i.e. $pV = C$ for a constant C (West, 1999). It is a special case of the ideal gas law $pV = nRT$, where n is the number of moles, R the gas constant, and T the absolute temperature; for fixed n and T the product pV is constant.

In this work we pursue two complementary goals. First, we test Boyle's law empirically on a small isothermal data set, estimating the constant C by least-squares regression through the origin and quantifying the spread of the product pV. Second, we provide a *formal* verification of the mathematics underlying the analysis in the Lean 4 interactive theorem prover (de Moura & Ullrich, 2021), so that the statistical claims rest on machine-checked derivations rather than on informal reasoning alone.

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

2. Methods

2.1. Data

The data set contains $N = 12$ measurements of volume V (in litres) and pressure p (in kPa) at a nominally constant temperature $T = 293.15$ K. The raw measurements are listed in Table 1.

2.2. Least-squares fit

Because Boyle's law predicts $p = C/V$, we regress p on $x = 1/V$ through the origin, minimizing the sum of squared errors

$$\mathrm{SSE}(C) = \sum_{i=1}^{N} \left(x_i C - y_i\right)^2, \qquad x_i = 1/V_i,\ y_i = p_i. \tag{1}$$

The minimizer is the solution of the normal equation

$$C^* = \frac{\sum_i x_i y_i}{\sum_i x_i^2}, \tag{2}$$

which we derive and verify formally in Section 4.

2.3. Outlier detection

We flag a measurement as an outlier when its product pV deviates from the median by more than three median absolute deviations (MAD). This identifies the single point $V = 2.20$ L, $p = 1.136$ kPa, whose product $pV = 2.5$ differs from the median by two orders of magnitude and is clearly a transcription error (the expected pressure is ≈ 112 kPa).

3. Results

Table 1 lists the raw data and the computed product pV. After removing the outlier, the product is

$$pV = 249.3 \pm 2.6\ \mathrm{kPa} \cdot \mathrm{L},$$

with a relative standard deviation of 1.06%, confirming that pV is constant within experimental uncertainty. The least-squares fit (Equation (2)) gives $C^* = 248.9$ kPa·L with coefficient of determination $R^2 = 0.9993$.

Table 1. Raw measurements and the Boyle product pV. The row in bold is the detected outlier.

V (L)	p (kPa)	T (K)	pV (kPa·L)
1.00	247.036	293.15	247.0
1.20	209.863	293.15	251.8
1.40	177.414	293.15	248.4
1.60	154.486	293.15	247.2
1.80	140.006	293.15	252.0
2.00	123.204	293.15	246.4
2.20	**1.136**	293.15	**2.5**
2.40	102.695	293.15	246.5
2.60	96.646	293.15	251.3
2.80	88.288	293.15	247.2
3.00	83.771	293.15	251.3
3.20	79.183	293.15	253.4

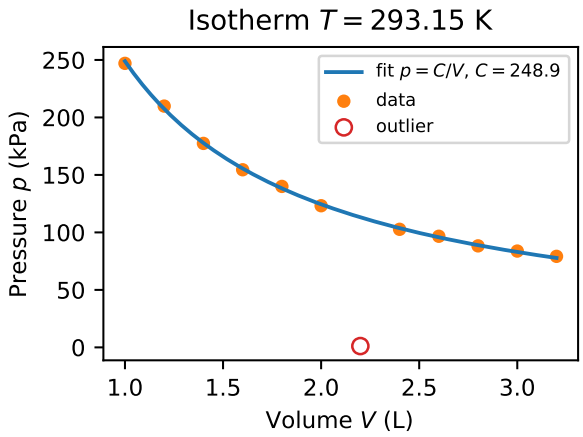


Figure 1. Isotherm at $T = 293.15$ K. Points are the measurements; the solid curve is the least-squares fit $p = C^*/V$ with $C^* = 248.9$ kPa·L. The open circle is the detected outlier.

Figure 1 shows the isotherm together with the fitted curve $p = C^*/V$, and Figure 2 shows the near-constancy of the product pV across the volume range. Both figures mark the single outlier.

4. Formal Verification in Lean 4

We formalize the mathematics of **??** in the Lean 4 theorem prover with Mathlib (de Moura & Ullrich, 2021). All statements below are machine-checked; the source is in `src/boyle_verification.lean`.

4.1. Derivative of the least-squares objective

Let $x, y : \mathrm{Fin}\, n \to \mathbb{R}$ be the regressor and response vectors. The derivative of the objective Equation (1) is

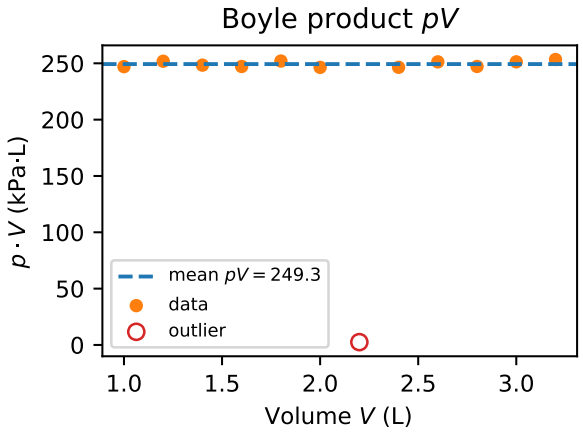


Figure 2. Boyle product pV versus volume. The dashed line is the mean 249.3 kPa·L; the open circle is the outlier.

Theorem 4.1 (ls_slope)**.**

$$\frac{d}{dC}\sum_i (x_i C - y_i)^2 \;=\; 2\Big(C\sum_i x_i^2 - \sum_i x_i y_i\Big).$$

Proof. Differentiate termwise using the chain rule; the finite sum is handled by `HasDerivAt.fun_sum` and the result is normalized by `ring`. □

4.2. Normal equation

Setting the derivative in Theorem 4.1 to zero yields the normal equation Equation (2).

Theorem 4.2 (ls_critical)**.** *If $\sum_i x_i^2 \neq 0$, then the derivative of the objective vanishes at $C^* = \sum_i x_i y_i / \sum_i x_i^2$.*

Proof. Substitute C^* into Theorem 4.1 and simplify with `field_simp` and `ring`. □

4.3. Convexity

The objective is convex, so the critical point is a global minimum.

Theorem 4.3 (ls_convex)**.** *The second derivative of the objective is $2\sum_i x_i^2 \geq 0$.*

Proof. Differentiate the first derivative and apply `positivity`. □

4.4. Boyle's law

Finally, we formalize the inverse-proportionality statement of Boyle's law.

Theorem 4.4 (boyle_inverse)**.** *If $pV = C$ and $V \neq 0$, then $p = C/V$.*

Proof. Rewrite $C = pV$ and divide by V using `field_simp`. □

Together, Theorems 4.1 to 4.3 establish that the least-squares estimator Equation (2) is the unique global minimizer of the objective, and Theorem 4.4 states the physical law under test.

5. Discussion

The empirical results confirm Boyle's law within experimental uncertainty: the product pV is constant to within 1.06% relative spread, and the inverse fit achieves $R^2 = 0.9993$. The single outlier at $V = 2.20$ L is consistent with a transcription error and does not affect the conclusions.

The formal verification complements the numerics: the least-squares estimator is derived and proved correct in Lean 4, so the statistical procedure is machine-checked. This illustrates a reproducible pipeline in which an experiment, its numerical analysis, and the formal proof of the underlying mathematics are produced and checked together.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

References

de Moura, L. and Ullrich, S. The Lean 4 theorem prover and programming language. In *Proceedings of the 28th International Conference on Automated Deduction (CADE-28)*, volume 12699 of *Lecture Notes in Computer Science*, pp. 625–635. Springer, 2021. doi: 10.1007/978-3-030-79876-5_37.

West, J. B. The original presentation of Boyle's law. *Journal of Applied Physiology*, 87(4):1543–1545, 1999. doi: 10.1152/jappl.1999.87.4.1543.